Kiana Sherafati

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Education

Amirkabir University of Technology Sept. 2022 – Present
B.Sc. in Computer Engineering | Tehran, Iran

- **GPA: 17.77/20.00.**
- **137 of 140 credits completed**; the undergraduate thesis is the sole remaining degree requirement.

Research Experience

ERP Text-to-SQL with Retrieval-Augmented Generation Oct. 2026
Undergraduate Thesis | Amirkabir University of Technology
Supervisor: Dr. Masoumeh Taromi Rad | Public repository

- Designed and implemented a retrieval-augmented pipeline translating English ERP reporting questions into Microsoft SQL Server T-SQL, addressing abbreviated identifiers and complex schema relationships.
- Combined BGE-M3 dense retrieval with MiniLM semantic/lexical retrieval and foreign-key expansion; merged candidates and reranked them with BGE-Reranker-v2-M3 before Qwen2.5-Coder-7B-Instruct generation.
- Evaluated schema retrieval and analyzed query-generation errors; reranking increased macro recall at 20 tables from **75.27% to 80.77%** in the thesis experiments.
- Implemented schema-aware SQL validation and parameterized, read-only previews through a Streamlit interface. Released reusable components and a synthetic demo while excluding company schemas and connection details.

Professional Experience

Green Dataware (Sabz Dadeh) Tehran, Iran
ERP Specialist (part-time) Sept. 2025 – Present
Previously: ERP/BI Software Engineering Intern, Summer 2025

- Developed SSRS organizational reports with dynamic tables and print-ready layouts, and an HR dashboard with drill-down views of headcount, contracts, and organizational units.
- Implemented ERP business logic using T-SQL and stored procedures; optimized queries and analyzed relationships across ERP modules.
- Diagnosed system issues using SQL Server Profiler and participated in client requirements analysis and product demonstrations.

Honors & Distinctions

Iranian Nationwide University Entrance Examination 2022
Entrance examination rank: **163**.

Gold and Bronze Medals in Collegiate Swimming 2024
Individual and relay events, 16th Cultural and Sports Olympiad for Students of Iranian Universities.

Selected Academic Projects

MyTorch: Neural Network Framework Spring 2025

Python, NumPy | *Source code*

- Developed an educational neural network framework with tensor automatic differentiation, backpropagation, dense layers, activation functions, and classification/regression losses.
- Implemented SGD, momentum, RMSProp, and Adam optimizers; applied an MLP to the Breast Cancer Wisconsin (Diagnostic) dataset.

AI Fundamentals: Search, Learning and Inference

Python, NumPy, Gymnasium | *Source code and reports*

- Implemented state-space search for Sokoban, AC-3 and heuristic backtracking for crossword constraints, and adversarial game search within course-provided environments.
- Explored policy iteration and tabular Q-learning with discretized LunarLander observations; implemented exact Bayesian inference and sampling methods, with reports documenting the course experiments.

AUJava-to-C Compiler

Java, ANTLR4, Maven, C | *Source code*

- Built a three-pass compiler for a Java subset, covering parsing, symbol collection, inheritance resolution, semantic type checking, lexical scoping, and C code generation.
- Implemented dynamic method dispatch using C structures and function pointers; included 27 positive and negative test cases for generated-program behavior and semantic error rejection.

Teaching Assistant Experience

Amirkabir University of Technology

Winter 2026	**Instructor**
Data Mining	Dr. E. Nazerfard
Software Engineering II	Dr. F. Gohari
Fall 2025	
Business Intelligence and Analysis **(Head Teaching Assistant)**	Dr. A. Ahmadi
Microprocessor and Assembly Language	Dr. H. Farbeh
Applied Linear Algebra	Dr. M. Chehreghani
Theory of Machines and Languages	Dr. M. Zakeri
Software Engineering I	Dr. E. Alirezaei
Fall 2024	
Applied Linear Algebra	Dr. E. Nazerfard
Principles of Database Design	Dr. Z. Poorbahman

Technical Skills

Programming: Python, SQL/T-SQL, C, C++, Java, JavaScript

Machine learning & data analysis: PyTorch, NumPy, pandas, scikit-learn; neural networks, reinforcement learning, Bayesian inference

LLMs & retrieval: Transformers, SentenceTransformers, FAISS, retrieval-augmented generation, hybrid schema retrieval, reranking

Databases & business intelligence: SQL Server, MySQL, SQLite, SSRS, Power BI

Software development: Django, Streamlit, SQLGlot, pyodbc, Git, Docker; object-oriented programming, design patterns, microservices, unit testing

Other: ANTLR4, Maven, HTML/CSS, Verilog, VHDL, LaTeX

Languages

Persian (native); English (advanced).